

Dylan Edwards

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EDUCATION

University of Illinois Urbana-Champaign

Graduated: May 2025

Bachelor of Science in Systems Engineering and Design

Minor in Computer Science

SKILLS

Technical Skills: FDM Printing, Laser Cutting, Silicon Molding, CNC Machining, PCB Design

Software Skills: Autodesk (Fusion 360, Inventor), OnShape, SolidWorks, Java, Python, C++, ROS

Research Skills: Data Collection & Analysis, Technical Documentation, Robotics & Automation

PROFESSIONAL EXPERIENCE

Robo-Touch Lab

Champaign, IL

Research Assistant

Aug. 2025 – Present

- Designed a robotic hand with direct-drive worm gear joints to mimic human motion
- Modeled hand kinematics in simulation to assess manipulability and reachability volume
- Developed evaluation framework for joint accuracy, power grasp forces, and repeatability testing

US Army Construction Engineering Research Laboratory

Champaign, IL

Undergraduate Research Assistant

Jan. 2024 – Jul. 2024

- Developed a waterproof electronic box featuring custom-designed PCBs, cables, and housing
- Designed a belt tensioning module with adjustability for different applications
- Created a component diagram illustrating internal and external connections within the box

Seibel Center for Design

Champaign, IL

Shop Attendant

Aug. 2023 – Jan. 2024

- Gained hands-on experience for CNC Machining, Laser Cutting, Waterjet Cutting, and Soldering
- Trained guests on the operation and file setup for the CNC and waterjet machines
- Consulted with guests to recommend the most efficient way of manufacturing their projects

Summer Intern

Jun. 2023 – Aug. 2023

- Increased process efficiency by developing manufacturing jigs for more consistency and accuracy
- Redesigned and implemented a new workspace layout using human-centered design principles
- Presented progress reports to clients to communicate important updates on the project

PROJECTS

Pneumatic Testing Apparatus for Crandall Stats & Sensors

Senior Design Project

- Created modular PLC-based software architecture for the automated quality control processes
- Performed market research on advanced flow sensors to expand system capabilities by over 50%

Thermal Eye

Mottier Innovation Challenge

- Developed equipment to aid firefighters in determining danger points within burning structures
- Designed a mount for a thermal camera on a NFPA standard helmet for hands-free scanning

LEADERSHIP EXPERIENCE

VEX Robotics

Champaign, IL

Mechanical Team Lead/Mentor

Oct. 2021 – Dec. 2024

- Managed and mentored a team of 15+ members to design and build 2 competition robots
- Fostered an inclusive design process that encouraged diverse perspectives
- Developed and implemented design techniques to improve team workflow and product quality

Outreach Director

Aug. 2023 – May 2024

- Organized tournaments for collegiate and high school students held at the university
- Collaborated with RSO's to create cad challenges to increase members' CAD experience
- Established a welcoming environment for members through social events within the club